A Brief Glossary of Notations

Sets

$\mathbb{R}$	real numbers
$\mathbb{C}$	complex numbers
$\cup$	set union
$\cap$	set intersection
$\varnothing$	empty set
$\in$	is a member of; belongs to
$\notin$	is not a member of; does not belong to
$\subset$	is a subset of
$\not\subset$	is not a subset of
$\forall$	for all; for each

Various Integers

n_{np}	number of nodal points
n_{en}	number of element nodes
n_{eq}	number of equations
n_{ee}	number of element equations
n_{dof}	number of degrees of freedom
n_{ed}	number of element degrees of freedom
n_{el}	number of elements
A, B	global node numbers $(1 \leq A, B \leq n_{np})$
a, b	element node numbers $(1 \leq a, b \leq n_{en})$
P, Q	global equation numbers $(1 \leq P, Q \leq n_{eq})$
p, q	element equation numbers $(1 \leq p, q \leq n_{ee})$
e	element number $(1 \leq e \leq n_{el})$

Element Arrays

m^e	mass matrix of the eth element
c^e	damping matrix of the eth element

k^e	stiffness matrix of the eth element
f^e	force vector of the eth element
a^e	acceleration vector of the eth element
v^e	velocity vector of the eth element
d^e	displacement vector of the eth element

Global Arrays

M	mass matrix
C	damping matrix
K	stiffness matrix
F	force vector
a	acceleration vector
v	velocity vector
d	displacement vector
$\mathbf{A}$	finite element assembly operator $\left(\text{e.g., } K = \mathop{\mathbf{A}}_{e=1}^{n_{el}} k^e\right)$

Boundary-Value Problems

n_{sd}	number of space dimensions
i, j, k, l	spatial indices $(1 \leq i, j, k, l \leq n_{sd})$
$\mathbb{R}^{n_{sd}}$	Euclidean n_{sd}-space
Ω	domain in $\mathbb{R}^{n_{sd}}$
$\bar{\Omega}$	closure of Ω
x_i, x	point in $\bar{\Omega}$
Γ	boundary of Ω
n_i, n	unit outward normal vector to Γ
Γ_g, Γ_{g_i}	part of the boundary where Dirichlet conditions are specified
Γ_h, Γ_{h_i}	part of the boundary where Neumann conditions are specified
δ_{ij}	Kronecker delta
η	set of node numbers
η_g, η_{g_i}	set of node numbers at which Dirichlet conditions are specified

Variational Methods

h	mesh parameter
$\mathcal{S}, \mathcal{S}_i, \mathcal{S}$	collections of trial solutions
$\mathcal{V}, \mathcal{V}_i, \mathcal{V}$	collections of weighting functions
$\mathcal{S}^h, \mathcal{S}_i^h, \mathcal{S}^h$	finite-dimensional collections of trial solutions
$\mathcal{V}^h, \mathcal{V}_i^h, \mathcal{V}^h$	finite-dimensional collections of weighting functions
u, u_i, u	trial solutions
w, w_i, w	weighting functions
u^h, u_i^h, u^h	finite-dimensional trial solutions
w^h, w_i^h, w^h	finite-dimensional weighting functions
ℓ, ℓ_i, ℓ	source terms
g, g_i, g	Dirichlet boundary data
h, h_i, h	Neumann boundary data
g^h, g_i^h, g^h	extensions of g, g_i, g, respectively, to Ω
e, e_i, e	errors
$L_2(\Omega)$	Hilbert space of square-integrable functions

$H^s(\Omega)$	Sobolev space of functions with s square-integrable generalized derivatives
$\|\cdot\|$	L_2 norm on Ω
$\|\cdot\|_s$	H^s norm on Ω
$a(\cdot, \cdot)$	symmetric bilinear form; strain-energy inner product
$(\cdot, \cdot)$	symmetric bilinear form; L_2 inner product on Ω
$(\cdot, \cdot)_\Gamma$	symmetric bilinear form; L_2 inner product on Γ

Data Processing Arrays

IEN (a, e)	element nodes matrix
ID (i, A)	destination matrix
LM (i, a, e)	location matrix

Heat Conduction

q_i	heat flux vector
κ_{ij}	conductivities
u	temperature
$u_{,i}$	temperature gradient
ℓ	prescribed heat supply
g	prescribed boundary temperature
h	prescribed boundary heat flux
d_A, d_a^e	nodal temperature
g_A, g_a^e	prescribed nodal temperature

Elasticity

ϵ_{ij}	infinitesimal strain tensor
σ_{ij}	Cauchy stress tensor
c_{ijkl}	elastic coefficients
u_i	displacement vector
$u_{(i,j)}$	symmetric part of the displacement gradients (equals ϵ_{ij})
ℓ_i	prescribed body force vector
g_i	prescribed boundary displacement vector
h_i	prescribed boundary traction vector
d_{iA}, d_{ia}^e	nodal displacements
g_{iA}, g_{ia}^e	prescribed nodal displacements
B	bulk modulus
E	Young's modulus
ν	Poisson's ratio
G, μ	shear modulus
λ	Lamé modulus
B, B_A, B_a	strain-displacement matrices
D, $\tilde{D}$	material moduli matrices

Isoparametric Elements

Ω^e	element domain $\subset \Omega$
Γ^e	boundary of Ω^e
x_1, x_2, x_3, x, y, z	Cartesian coordinates
$\square$	parent domain of a quadrilateral or hexahedral element
$\boldsymbol{x}^e: \square \to \bar{\Omega}^e$	isoparametric mapping
ξ, η	element natural coordinates for quadrilaterals
ξ, η, ζ	element natural coordinates for hexahedra
$\boldsymbol{\xi}$	point in $\square$
j	Jacobian determinant of the mapping $\boldsymbol{x}^e(\boldsymbol{\xi})$

$C^0(\Omega)$	the class of continuous functions
$C^k(\Omega)$	the class of continuous functions possessing k continuous derivatives
N_A, N_a^e	shape functions
x_A, x_a^e	nodal points
r, s, t	triangular coordinates; area coordinates
r, s, t, u	tetrahedral coordinates; volume coordinates
$l_a(\xi)$	Lagrange polynomial associated with the ath element node
$\tilde{\xi}_l$	location of the lth integration point
W_l	weight assigned to lth integration point
$U1$, $U2$, $U3$	uniform integration elements
$S1$, $S2$, $S3$	selective integration elements

Incompressible and Nearly-Incompressible Elasticity

$\mathcal{P}$	pressure trial solution and weighting function space
$\mathcal{P}^h$	finite-dimensional pressure trial solution and weighting function space
$p, q \in \mathcal{P}$	trial pressure and weighting function
$p^h, q^h \in \mathcal{P}^h$	finite-dimensional trial pressure and weighting function
g^e, G	element and global gradient matrices
$(g^e)^T$, G^T	element and global divergence matrices
p	vector of nodal pressures
B_a^{dev}	deviatoric strain-displacement matr
B_a^{dil}	dilatational strain-displacement matr
$\bar{B}_a^{\text{dil}}$	improved dilatatio strain-displacemen matrix
$\bar{B}_a$	improved strain-displacement matri

Plates

$\alpha, \beta, \gamma, \delta$	tensor indices ($1 \leq \alpha, \beta, \gamma, \delta \leq$
$e_{\alpha\beta}$	alternator tensor
w	transverse displacement
θ_α	rotation vector
$\kappa_{\alpha\beta} = \theta_{(\alpha,\beta)}$	curvature tensor
γ_α	shear strain vector
$c_{\alpha\beta\gamma\delta}$, $c_{\alpha\beta}$	elastic coefficients
$m_{\alpha\beta}$	moment tensor
q_α	shear force vector
W	prescribed boundary displacement
Θ_α	prescribed boundary rotations
F	total applied transverse force
C_α	total applied couple
M_α	prescribed boundary moments
Q	prescribed boundary shear force
s	arc-length paramete of plate boundary
$\hat{\theta}_\alpha = e_{\alpha\beta}\theta_\beta$, $\hat{\theta}_s$, $\hat{\theta}_n$	right-hand-rule rotations
k_b^e, k_s^e	eth element bending and shear stiffness

D^b, D^s	matrices of bending and shear moduli
B^b, B^s	bending and shear strain-displacement matrices

Beams

w_i	displacements
θ_i	rotations
κ_α	curvatures
γ_α	shear strains
ϵ	axial strain
ψ	twist
m_i	moments
q_i	shear forces
W_i	prescribed end displacements
Θ_i	prescribed end rotations
F_i	total applied force per unit length
C_i	total applied couple per unit length
M_i	prescribed end moments
Q_i	prescribed end shears
A	cross-sectional area
A_α	shear areas
I_α	moments of inertia
J	polar moment of inertia
k_a^e, k_b^e, k_s^e, k_t^e	eth element axial, bending, shear, and torsional stiffness, respectively

Shells

x	position vector
$\bar{x}$, $\bar{x}_a$	position vectors to references surface
X, $\hat{X}_a$	fiber vectors
$z_a(\zeta)$	thickness function
e_1^l, e_2^l, e_3^l	lamina basis vectors
e_1^f, e_2^f, e_3^f	fiber basis vectors
u	displacement vector
$\bar{u}$, $\bar{u}_a$	displacement of reference surface
U, U_a, $\hat{U}_a$	fiber displacements
$\tilde{\sigma}^l = \tilde{D}^l \tilde{\epsilon}^l$	reduced constitutive equation in the lamina basis
$m_{\alpha\beta}$	moments
n_α	membrane forces
q_α	transverse shear forces

Dynamics

t	time
Δt	time step
n	time step number
a_n, v_n, d_n	approximations of $a(t_n)$, $v(t_n)$, and $d(t_n)$, respectively
$\tilde{a}_n$, $\tilde{v}_n$, $\tilde{d}_n$	predictor values of acceleration, velocity, and displacement, respectively
Δt_{crit}	critical time step
A	amplification factor
$\boldsymbol{A}$	amplification matrix
$\rho(A)$	spectral radius of amplification matrix
τ, $\boldsymbol{\tau}$	local truncation errors
λ, λ^h, $\bar{\lambda}^h$	eigenvalues
ψ, ϕ	eigenvectors
ω, ω^h, $\bar{\omega}^h$	frequencies
$\bar{\xi}$	algorithmic damping ratio
$(\bar{T} - T)/T$	relative period error
Ω_{crit}	critical sampling frequency

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$\boldsymbol{M}^I, \boldsymbol{C}^I, \boldsymbol{K}^I, \boldsymbol{F}^I$	implicit mesh partition of mass, damping, stiffness, and force
$\boldsymbol{M}^E, \boldsymbol{C}^E, \boldsymbol{K}^E, \boldsymbol{F}^E$	explicit mesh partition of mass, damping, stiffness, and force
$u_{0i}, \dot{u}_{0i}$	initial displacem and velocities
u_0	initial temperatu
ρ	density
c	capacity